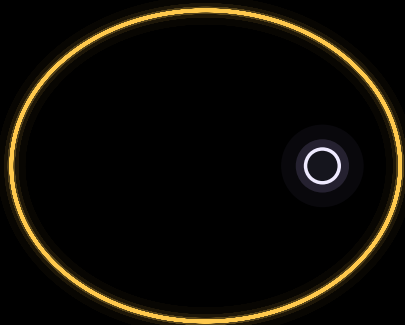


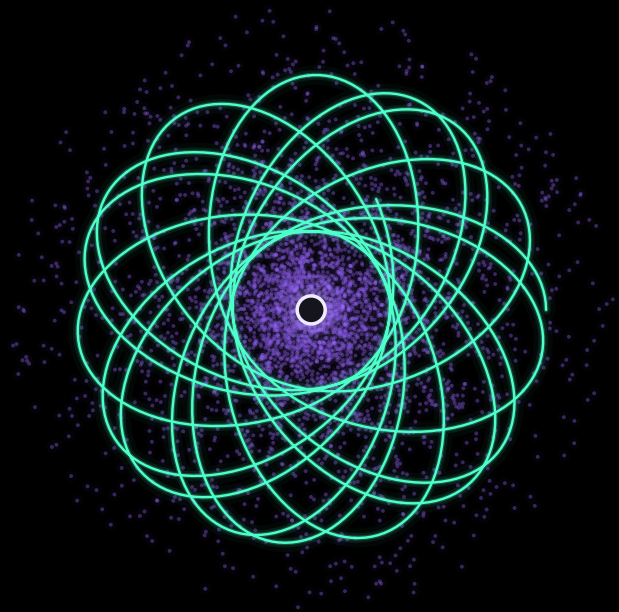
Figure 11. Three kinds of orbit in the pulled grid (computed; seen from above)

(a) Elliptical orbit around a lone mass



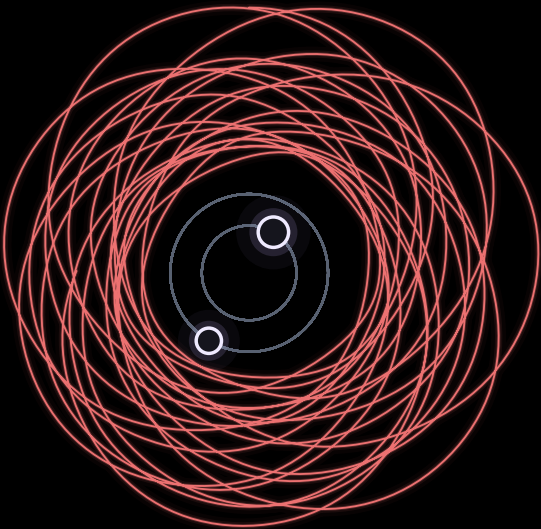
five laps drawn: they land exactly on top of each other. The mass sits at one focus; the body speeds up when close and slows when far (Kepler's laws).

(b) Irregular orbit inside the fluid pool



the pull grows as the body dips deeper into the fluid, so each lap swings round further: the orbit never closes and fills a ring (how stars move in galaxies).

(c) Irregular orbit near two masses



two masses circle each other (grey); a light body is tugged by both and never repeats. Shift its start by one part in a million and the gap grows $\sim 1,629$ -fold.

Equations: $\ddot{\mathbf{x}} = -\nabla\phi$. (a) $\phi = -GM/r$; $r = a(1 - e^2)/(1 + e\cos\theta)$, $T^2 = 4\pi^2 a^3/GM$. (b) $M(r) = M + M_h r^3/(r^2 + r_c^2)^{3/2}$, $\ddot{\mathbf{x}} = -GM(r)\hat{r}/r^2$. (c) $\phi = -\sum_k Gm_k/|\mathbf{x} - \mathbf{x}_k(t)|$; gap $\delta(t) \approx \delta_0 e^{\lambda t}$.